

# IRC Checklist & Application

PS Ref: PS-REF

## Combined IRC Checklist Template

| Check ID                                                            | Type        | Check Description                         | Example Symptom                                                                      |
|---------------------------------------------------------------------|-------------|-------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Imperative Checks (Obligation Strength &amp; Responsibility)</b> |             |                                           |                                                                                      |
| IMP-1                                                               | Imperative  | Priority vs. wording mismatch             | “The system should maintain connectivity” (Essential requirement with weak language) |
| IMP-2                                                               | Imperative  | Not written as clear obligation           | “Emergency handling is important for mission safety” (Descriptive, not imperative)   |
| IMP-3                                                               | Imperative  | Responsibility unclear                    | “Mission data must be protected” (Ambiguous who protects it)                         |
| <b>Continuance Checks (Complexity &amp; Compound Statements)</b>    |             |                                           |                                                                                      |
| CON-1                                                               | Continuance | Compound requirement                      | “System shall log, store, and analyze mission data” (Three requirements combined)    |
| CON-2                                                               | Continuance | Deferred reference dependency             | “As defined in the following section” (Meaning depends on undefined content)         |
| <b>Contextual Checks (Meaning, Testability &amp; Consistency)</b>   |             |                                           |                                                                                      |
| CTX-1                                                               | Contextual  | Ambiguous term / multiple interpretations | “Sufficient bandwidth”, “Safe operational state”, “User-friendly”                    |
| CTX-2                                                               | Contextual  | Not testable / not verifiable             | “System shall be reliable” (No measurable reliability metric)                        |
| CTX-3                                                               | Contextual  | Inconsistent / out of scope               | Conflicts with another requirement or outside project boundaries                     |

## IRC Application Format & Recording Template

When applying the IRC framework during the Breakdown activity, record findings systematically using the following format:

| PS Ref | IRC Flags | Reason                                        | Clarify? | Question for Client |
|--------|-----------|-----------------------------------------------|----------|---------------------|
| PS-REF | –         | No incongruities identified under IRC checks. | N        | N/A                 |

**Note:** The complete filled IRC application table with all project findings and resolution results is provided in the main Analysis Report.

## Flagged Requirements (10 PS Clauses)

From IRC Application Results

| PS Ref    | IRC Flags           | Reason                                                                                | Clarify? | Question for Client                                                                                        |
|-----------|---------------------|---------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------|
| BO-L1-4   | CTX-1, CTX-2, CON-2 | "Abnormal conditions" ambiguous and not testable; no definition.                      | Y        | What quantitative thresholds define "abnormal conditions" for pressure, temperature, and power deviations? |
| IT-L0-C2  | IMP-1, CTX-2        | Essential constraint with ambiguous language, no measurable parameters.               | Y        | What specific minimum bandwidth and latency constraints apply?                                             |
| IT-L0-C3  | CTX-1               | "Protected in accordance with applicable security obligations" – undefined standards. | Y        | Which specific security standards or regulatory frameworks must the system comply with?                    |
| BO-L1-5   | CTX-1, CON-1        | "Orderly manner" ambiguous; compound continuation/termination/recovery.               | Y        | What specific operational steps define an "orderly" shutdown and recovery process?                         |
| IT-L0-A1  | CTX-1, CON-1        | "Primary mission sensors supplied by customer" – missing interface protocols.         | Y        | What middleware, communication protocols, and data formats are required for sensor integration?            |
| BO-L0-A1  | CTX-3               | "Missions supervised by trained personnel" – out of scope for software.               | Y        | Should this supervision requirement be recorded as an operational assumption (not a software requirement)? |
| IT-L1-5   | CON-1               | "Operational Logging Capability" combines logging, storage, retrieval, analysis.      | N        | N/A                                                                                                        |
| BO-L1-6   | IMP-3               | "Operational Accountability" – missing clear responsibility assignment.               | Y        | Which system component is responsible for logging and attributing operator accountability?                 |
| BO-L2-4.1 | IMP-2               | "Abnormal operating conditions shall be identifiable" – not clear system obligation.  | N        | N/A                                                                                                        |
| BO-L2-4.2 | CTX-1               | "Safe operational state" – ambiguous safety condition definition.                     | Y        | What specific technical conditions define a "safe operational state" for the submersible?                  |

## Non-Flagged Requirements (17 PS Clauses)

No incongruities identified under IRC checks

| PS Ref    | IRC Flags | Reason                                        | Clarify? | Question for Client |
|-----------|-----------|-----------------------------------------------|----------|---------------------|
| BO-L0-C1  | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L0-C2  | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L0-C3  | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L0-C4  | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L1-1   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L1-2   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L1-3   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L1-7   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L2-6.1 | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| BO-L2-6.2 | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L0-C1  | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L1-1   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L1-2   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L1-3   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L1-4   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L1-6   | –         | No incongruities identified under IRC checks. | N        | N/A                 |
| IT-L2-5.1 | –         | No incongruities identified under IRC checks. | N        | N/A                 |